
DP-DRO algorithm design

Distributionally Robust Learning with Differential Privacy

Shruthi Kannappan

Department of Computer Science
Indian Institute of Technology Madras
Chennai - 600036
cs21b077@ssmail.iitm.ac.in

Sushanth Reddy

Department of Electrical Engineering
Indian Institute of Technology Madras
Chennai - 600036
ee21b102@ssmail.iitm.ac.in

Abstract

Distributionally Robust Optimization (DRO) improves fairness by focusing on minority groups, while Differential Privacy (DP) protects individual data points. Balancing fairness and privacy is a key challenge in developing ethical and deployable models. Below, we implement the DRO and Spectral Risks algorithms with Differential Privacy, for which we derive tight theoretical bounds on sensitivity. We benchmark the performance of these algorithms with DP across various fairness metrics.

1 Introduction

Distributional Robustness in machine learning models can be conceptualized as a measure of how well a model handles rare cases in the training dataset. Traditional models trained using Empirical Risk Minimization (ERM) tend to overlook minority groups (rare cases) as they focus on minimizing average loss. Distributionally Robust Optimization (DRO) addresses this issue by minimizing a loss function that assigns greater weight to these minority groups, typically those with the highest loss. Consequently, models trained using DRO often perform better on fairness metrics due to their increased attention to rare cases.

Distributionally Robust Learning modifies the training objective to minimize a different loss function:

$$R_D(\theta) := \max_{p \in P} \sum_{i=1}^n p_i l_i(\theta)$$

To regularize p and obtain sensitivity of the algorithm, we can minimize a log function of the form:

$$R_D(\theta) := \max_{p \in P} \sum_{i=1}^n p_i l_i(\theta) - \nu h(p)$$

where h is a strongly convex function.

Differential Privacy (DP) ensures that the inclusion or exclusion of any individual data point has a limited impact on the output of the model. A randomized algorithm $\mathcal{A}$ is said to be (ϵ, δ) -differentially private if, for any two datasets D and D' differing in a single data point, and for any possible output S of the algorithm, the following holds:

$$\Pr[\mathcal{A}(D) = S] \leq e^\epsilon \Pr[\mathcal{A}(D') = S] + \delta$$

Here, ϵ controls the privacy loss, and δ allows a small probability of failure. DP aims to minimize the influence of individual data points, ensuring privacy, but this may reduce the model’s sensitivity to minority group datapoints, potentially impacting fairness. While DP aims to minimize the impact of individual data points on the model’s output, thereby enhancing privacy, DRO seeks to optimize performance across worst-case distributions, often amplifying the influence of minority groups or rare cases. This fundamental conflict poses a significant challenge in the development of machine learning models that simultaneously achieve both fairness and privacy. Addressing this is a critical step towards developing ethically robust and practically deployable machine learning systems that can maintain individual privacy without sacrificing fairness across diverse population subgroups.

This project revolves around developing theoretical foundations and practical algorithms that effectively combine Distributionally Robust Optimization and Differential Privacy. Distributionally Robust Learning modifies the training objective to minimize a different target function:

$$R_D(\theta) := \max_{p \in \mathcal{P}} \sum_{i=1}^n p_i l_i(\theta)$$

Where:

- $p \in \mathbf{R}^n$: A vector of coefficients $p = (p_1, p_2, \dots, p_n)$.
- $l_D(\theta)$: The loss function: the loss values of predictions on a given Dataset D dependent on the parameter θ , $l_D(\theta) = (l_1(\theta), l_2(\theta) \dots l_n(\theta))$ where $l_i(\theta)$ is the loss of the i^{th} datapoint in D.
- ν : A scaling parameter for the regularization term.
- $h(p)$: A regularization term involving the vector $\mathcal{P}$ which is a strongly convex function.

To regularize p and obtain sensitivity of the algorithm, we can minimize a target function of the form:

$$R_D(\theta) := \max_{p \in \mathcal{P}} \sum_{i=1}^n p_i l_i(\theta) - \nu h(p)$$

where h is a strongly convex function.

The above objective can be minimized by algorithms like the Stochastic Subgradient Method (SGD). We developed a Differentially Private version of it by adding noise during a parameter update at each step/iteration of training. We proved a bound on the sensitivity in this algorithm and used it for noise addition. We then benchmark it against some common fairness metrics for various datasets and make some conclusions.

2 Literature Review/Related Work

Distributionally Robust Optimization (DRO) algorithms have been studied and developed in recent years. Sagawa et al. [2020] introduced DRO as a way to address uncertainties in data by optimizing for the worst-case distribution within an uncertainty set. Efficient DRO algorithms utilizing Spectral Risk Functions as weights have been developed by Mehta et al. [2022], where the trained models were evaluated on various fairness metrics. Specifically, the use of spectral risk functions allows for an interpolation between optimizing for average-case and worst-case performance, which is crucial for tasks where performance under uncertainty is important.

The Prospect model, proposed by Mehta et al. [2024], is a stochastic gradient-based algorithm that only requires tuning a single learning rate hyperparameter. They prove that it enjoys linear convergence for smooth, regularized losses, making it a more efficient alternative compared to previous methods that required tuning multiple hyperparameters.

These references have provided us with a strong understanding of DRO, helping us build intuition on its theory, implementation, and experimental evaluation. They have also served as a starting point for the development of **Differentially Private models**, where we aim to add noise to the DRO models to further enhance their fairness, especially for minority groups.

3 Sensitivity Calculations

3.1 Sensitivity Calculation for the generalized DP-DRO Algorithm

3.1.1 Objective Function

For addressing the DRO problem we are setting the model's aim to optimizing (towards minimum) a target function of the form for a given dataset D :

$$R_D(\theta) := \max_{p \in P} \sum_{i=1}^n p_i l_i(\theta) - \nu h(p)$$

The objective can be rewritten as $R_D(\theta) := \max_{p \in P} \phi_D(\theta, p)$ where $\phi_D(\theta, p) = p^T l_D(\theta) - \nu h(p)$. The maximization problem is formulated as:

$$R_D(\theta) = \max_{p \in P} \phi(\theta, p) = \max_{p \in P} (p^T l(\theta) - \nu h(p))$$

We define the set $\mathcal{P}$ as a subset of vector space $\mathbf{R}^n$ where we seek to optimize the objective function.

$$P = \{p \in \mathbb{R}^n : \sum_{i=1}^n p_i = 1, p_i \geq 0\}$$

The optimal weights for a given Dataset D is given by: $p_D^*(\theta) = \arg \max_{p \in P} \phi_D(\theta, p)$

3.1.2 Gradient Analysis

For a specific dataset D , this becomes:

$$\nabla R_D(\theta) = p_D^{*T} \nabla_{\theta} l_D(\theta)$$

(Refer Appendix 4.1 for derivation). For a dataset $D = (u_1, u_2, \dots, u_n)$, the loss function can be defined as:

$$l_D(\theta) = (loss(\theta, u_1), loss(\theta, u_2), \dots, loss(\theta, u_n))$$

The gradient of the loss function is:

$$\nabla_{\theta} l_D(\theta) = (\nabla_{\theta} loss(\theta, u_1), \nabla_{\theta} loss(\theta, u_2), \dots, \nabla_{\theta} loss(\theta, u_n))$$

Let us consider a neighboring dataset $D' = (u_1, u_2, \dots, u'_n)$, such that $loss(\theta, u'_n) = 0$ and $\nabla_{\theta} loss(\theta, u'_n) = 0$.

$$\begin{aligned} \Rightarrow l_{D'}(\theta) &= (loss(\theta, u_1), loss(\theta, u_2), \dots, loss(\theta, u_{n-1}), 0) \\ \Rightarrow \nabla_{\theta} l_{D'}(\theta) &= (\nabla_{\theta} loss(\theta, u_1), \nabla_{\theta} loss(\theta, u_2), \dots, \nabla_{\theta} loss(\theta, u_{n-1}), 0) \end{aligned}$$

3.1.3 Sensitivity Calculation

The sensitivity Δ is defined as follows:

$$\Delta = \max_{D \simeq D'} \|\nabla_{\theta} R_D(\theta) - \nabla_{\theta} R_{D'}(\theta)\|_2$$

For all datasets $D \simeq D'$:

$$\nabla_{\theta} R_D(\theta) - \nabla_{\theta} R_{D'}(\theta) = p_D^{*T} \nabla_{\theta} l_D(\theta) - p_{D'}^{*T} \nabla_{\theta} l_{D'}(\theta)$$

Here, $\nabla_{\theta} l_D(\theta)$ and $\nabla_{\theta} l_{D'}(\theta)$ differ at only one data point:

$$\nabla_{\theta} l_{D'}(\theta) = \nabla_{\theta} l_D(\theta) + \nabla_{\theta} l_{diff}(\theta)$$

where

$$l_{diff}(\theta) = l_{D'}(\theta) - l_D(\theta)$$

Then, we can express $\nabla_{\theta} R_{D'}(\theta)$ as follows:

$$\nabla_{\theta} R_{D'}(\theta) = p_{D'}^{*T} \nabla_{\theta} l_D(\theta) + p_{D'}^{*T} \nabla_{\theta} l_{diff}(\theta)$$

Subtracting these, we find:

$$\nabla_{\theta} R_{D'}(\theta) - \nabla_{\theta} R_D(\theta) = (p_{D'}^{*T} - p_D^{*T}) \nabla_{\theta} l_D(\theta) + p_{D'}^{*T} \nabla_{\theta} l_{diff}(\theta)$$

Using the Triangle Inequality, we have:

$$\|\nabla_{\theta} R_{D'}(\theta) - \nabla_{\theta} R_D(\theta)\|_2 \leq \|(p_{D'}^{*T} - p_D^{*T}) \nabla_{\theta} l_D(\theta)\|_2 + \|p_{D'}^{*T} \nabla_{\theta} l_{diff}(\theta)\|_2$$

Now, we analyze each term separately:

1. First term:

$$(p_{D'}^{*T} - p_D^{*T}) \nabla_{\theta} l_D(\theta) = \sum_{i=1}^n (p_{D_i}^* - p_{D'_i}^*) \nabla_{\theta} l_{D_i}(\theta)$$

Thus, we can bound it as follows:

$$\|(p_{D'}^{*T} - p_D^{*T}) \nabla_{\theta} l_D(\theta)\|_2 \leq \sum_{i=1}^n |p_{D_i}^* - p_{D'_i}^*| \|\nabla_{\theta} l_{D_i}(\theta)\|_2 \leq \sum_{i=1}^n |p_{D_i}^* - p_{D'_i}^*| C$$

Here, C represents a bound on the norm of the per-sample gradients, with $\|\nabla_{\theta} l_D(\theta)\|_2 \leq C$.

Therefore, we have:

$$\|(p_{D'}^{*T} - p_D^{*T}) \nabla_{\theta} l_D(\theta)\|_2 \leq \|p_D^* - p_{D'}^*\|_1 C$$

2. Second term:

$$\|p_{D'}^{*T} \nabla_{\theta} l_{diff}(\theta)\|_2 \leq \|p_{D'}^*\|_1 C$$

Since $p_{D'}^* \leq 1$, this can be simplified to:

$$\begin{aligned} \|p_{D'}^{*T} \nabla_{\theta} l_{diff}(\theta)\|_2 &\leq C \\ \Delta &\leq C (\|p_D^* - p_{D'}^*\|_1 + 1) \\ \|p_D^* - p_{D'}^*\|_1 &\leq \frac{B}{\nu} \quad (\text{Refer to Appendix 4.2 for derivation}) \\ \Rightarrow \|p_D^* - p_{D'}^*\|_1 &\leq \min(2, \frac{B}{\nu}) \quad (\text{Refer to Appendix 4.2 for derivation}) \end{aligned}$$

where B is defined as the maximum loss of a data point in the dataset:

$$B = \max l_D(\theta, u_n)$$

Finally, we conclude:

$$\Delta \leq \left(\min(2, \frac{B}{\nu}) + 1 \right) C$$

This result indicates that as the regularization constant increases, less noise is required to achieve differential privacy for the model. Similarly, for $h(p) = \frac{1}{2} \sum_{i=1}^n p_i^2$, we derive the same value of sensitivity.

3.2 Sensitivity for DP-SGD with spectral risk measures

In this method, p is a permutation of chosen fixed constants $\sigma_1 \leq \sigma_2 \leq \dots \leq \sigma_n$ where $\sum_{i=1}^n \sigma_i = 1$. In this case, the objective is obtained by taking the dot product of $\sigma = (\sigma_1, \sigma_2, \dots, \sigma_n)$ with $\text{sort}(l_D(\theta))$, where data points are sorted in ascending order of loss values. Hence $l_1(\theta) \leq l_2(\theta) \dots l_n(\theta)$. Using the same definition of adjacent datasets as in DRO, a

$$\begin{aligned} \nabla R_D(\theta) &= \sum_{i=1}^n \sigma_i \nabla l_{D_i}(\theta) \\ \nabla R_{D'}(\theta) &= \sum_{i=1}^{n-1} \sigma_{i+1} \nabla l_{D_i}(\theta) \\ \nabla R_D(\theta) - \nabla R_{D'}(\theta) &= \sum_{i=1}^{n-1} (\sigma_{i+1} - \sigma_i) \nabla l_{D_i}(\theta) + \sigma_n \nabla l_{D_n}(\theta) \\ \Delta &\leq \sum_{i=1}^{n-1} |\sigma_{i+1} - \sigma_i| \|\nabla l_{D_i}(\theta)\|_2 + |\sigma_n| \|\nabla l_{D_n}(\theta)\|_2 \leq C \left(\sum_{i=1}^{n-1} (\sigma_{i+1} - \sigma_i) + \sigma_n \right) \\ &\Rightarrow \Delta \leq C(2\sigma_n - \sigma_1) \end{aligned}$$

4 Experiments

4.1 Training Curve

Suboptimality measures the performance of an algorithm in terms of the difference between the achieved objective function value $F_\sigma(w)$ and the optimal value $F_\sigma(w^*)$. It is normalized by the initial value $F_\sigma(w^{(0)})$ to evaluate how close the solution is to the best possible outcome relative to the starting point. The suboptimality is given by the following formula:

$$\text{Suboptimality}(w) = \frac{F_\sigma(w) - F_\sigma(w^*)}{F_\sigma(w^{(0)}) - F_\sigma(w^*)}$$

where w^* is approximated by running LBFGS from Nocedal and Wright [1999] to convergence on the objective. The x-axis represents the number of calls to a first-order oracle (gradient and function value evaluations), normalized by the dataset size n . This metric helps assess how quickly the algorithm approaches optimality during training, accounting for different spectral risk objectives and datasets.

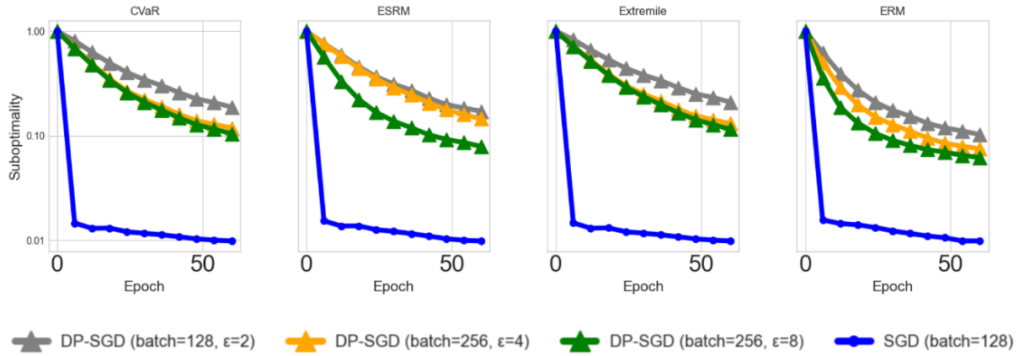


Figure 1: Variation of Sub Optimality for different Epsilons on ACS Income Dataset

We observe that the SGD algorithm converges much faster than DP-SGD algorithms for different ϵ values, even after tuning over different batch sizes. This is due to the addition of noise in DP-SGD algorithms. A higher ϵ ($\epsilon = 8$, less private since less noise is added) in DP-SGD converges faster than a lower ϵ ($\epsilon = 2$) algorithm. The effect of epsilon on sub-optimality is much more pronounced on steeper spectral risks measures (Extremile and CVaR) than ERM.

4.2 Fairness

Fairness metrics in AI are quantitative tools designed to measure bias in machine learning models, ensuring equitable treatment of individuals across different demographic groups regardless of sensitive attributes like race, gender, or age. The following are some of the common fairness metrics we explored and benchmarked our models against.

1. **Demographic Parity** Demographic parity is a metric that quantifies the independence of the models output with respect to membership in a demographic group. This means that low demographic parity implies that selection rates for different demographic groups is equal.

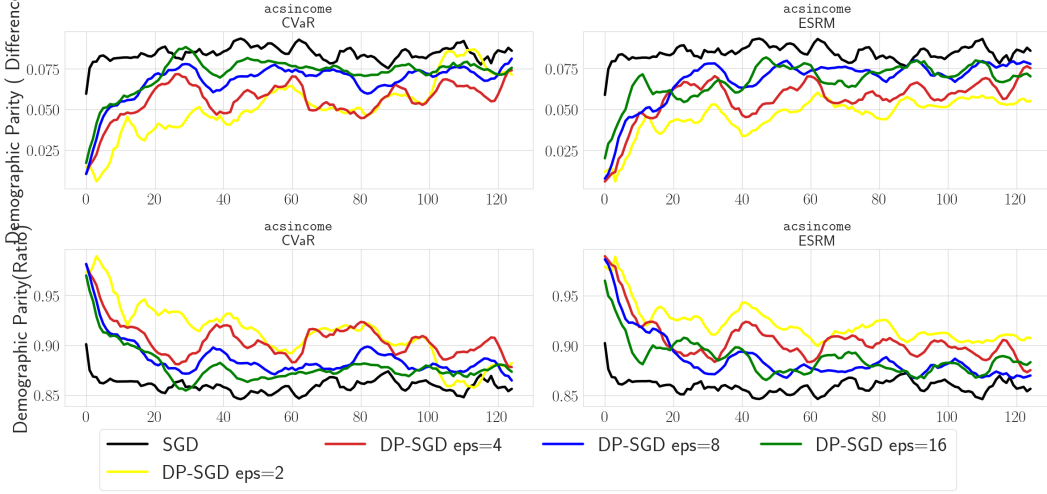


Figure 2: Demographic parity with Gender as Sensitive group

From 2, we can see that demographic parity is lower for private models, and the trend is that it decreases with lower epsilons. This makes intuitive sense as differential privacy aims to decrease the effect of one datapoint on the models output and demographic parity measures the effect of membership in one demographic group on the models output. So more private models tend to have lower demographic parity.

2. **The Equalized Odds** The goal of the equalized odds fairness metric is to ensure a machine learning model performs equally well for different groups. It is stricter than demographic parity because it requires that the machine learning model's predictions are not only independent of sensitive group membership, but that groups have the same false positive rates and true positive rates.

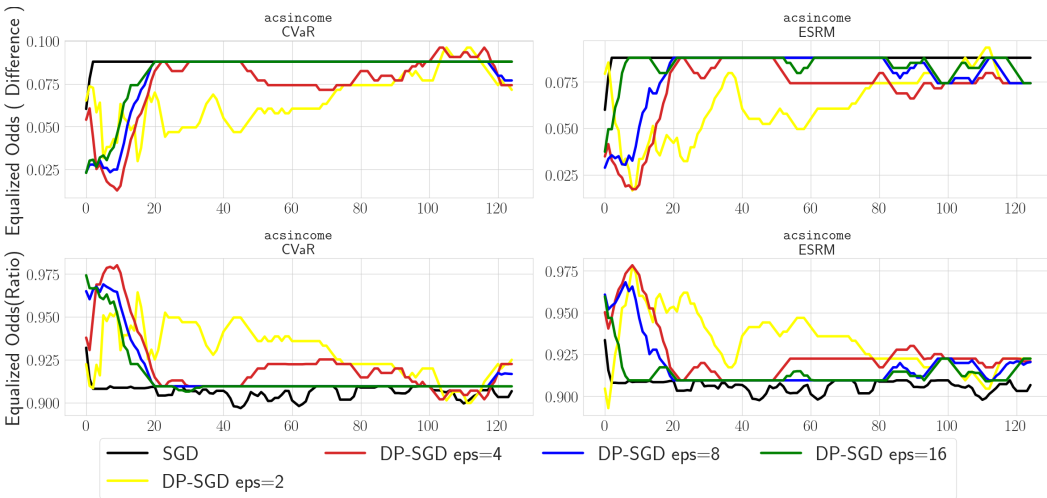


Figure 3: Equalized Odds with Gender as Sensitive group

From 3 we can indeed see that Equalized odds is more difficult to achieve however private models converge to a value slightly better than the non-private model.

3. **Statistical parity(Dist groups)** measures the maximum deviation between the distributions for any over any sensitive attribute. It is a notion of measuring distance between the distributions of outputs for sensitive groups.

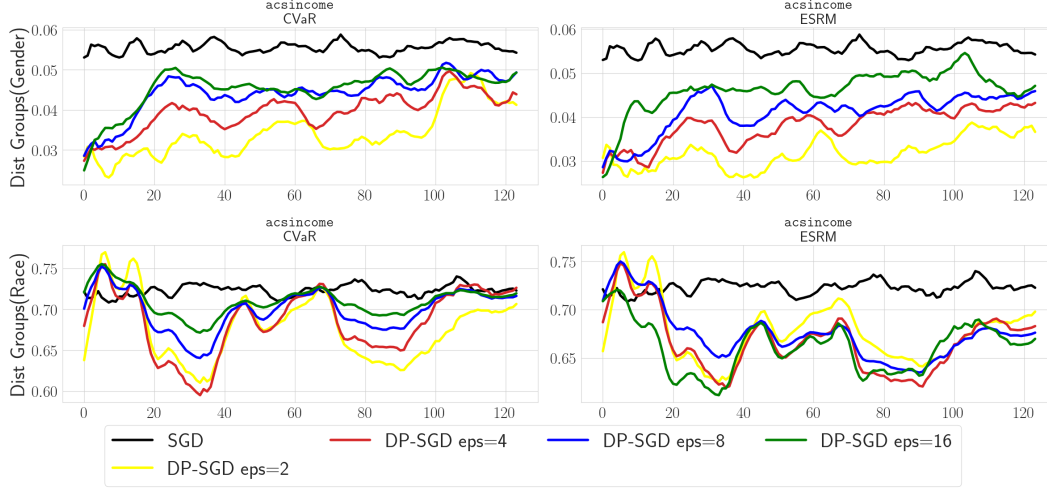


Figure 4: Distance between groups

From 4 we can see that for the Sensitive group gender, the distributions are closer for more private models. This is intuitive since the aim of differential privacy is to reduce the distance between distributions of neighboring datasets. The difference is quite visible in gender group compared to the race group.

4.3 Inferences

From the Fairness plots of private models for various epsilons and non-private model we can conclude the following

- The notion of fairness tries to reduce the bias in certain aspects towards or against demographic groups
- Membership of a point in a minority group leaks more information about it than a point in a majority group. Privacy aims to reduce this leakage. Hence the model tends to not learn these biases which give away the information of that point.
- Since the Distributionally Robust models learn the minority groups better, they tend to leak more information about the minority groups and show bias towards or against them.
- Differential privacy works against this privacy leakage and hence reduce the bias, making more fair for minority groups.

4.4 Implementation

Implemented the Differentially Private version of Sub-Gradient Descent Algorithm for Spectral Risk Measures from Mehta et al. [2022]. The noise multiplier is obtained using the DP-accounting library for a given ϵ and δ . Here we can use the dp-accounting library to compose over each step since we are privatising the gradients at each step equally.

5 Conclusion

This project spans from proving theoretical proofs to implementation of differentially private DRO models. We observe that there is a considerable trade off between optimality and privacy for different

spectral risk measures. The evaluation on several fairness metrics lead us to an interesting relation between privacy and fairness. We find that there is an added benefit of decreasing demographic group biases by making our models private. This could be of importance in today's and upcoming models as there are minority groups and a need to preserve privacy at the same time across the world.

6 Future Work

The future work will focus on several key areas:

- **Sensitivity Bound for LSVRG with Differential Privacy:** Derive the sensitivity bound for the LSVRG and other better performing algorithm when combined with differential privacy (DP). This will help ensure the algorithm's robustness while maintaining privacy guarantees at reasonable accuracies.
- **Implementation of Differentially Private DRO Algorithms:** Implement Differentially Private Distributionally Robust Optimization (DP-DRO) and DP-DRO Wessetrian algorithms, and benchmark their performance across various fairness metrics.
- **Exploration of Fairness-Privacy Relationship:** Experiment rigorously with and explore the theoretical relationship between fairness and privacy, aiming to understand how these critical aspects can be balanced and optimized in machine learning systems.

7 Other Experiments

7.1 Suboptimality for Epsilon 1024 on ACS Income Dataset

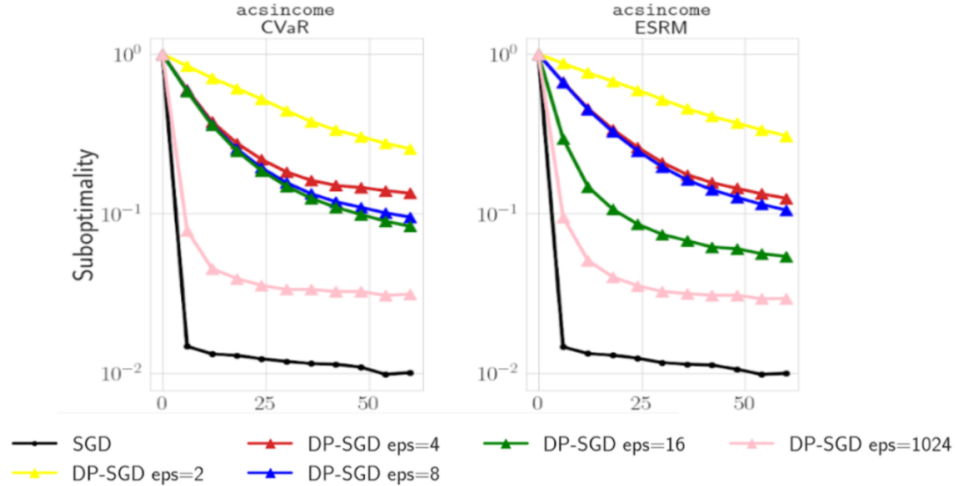


Figure 5: Suboptimality for Epsilon 1024 on ACS Income Dataset

Observation: The suboptimality of DP-SGD with $\epsilon=1024$, is not equal to that of SGD, even though the noise added is almost zero.

Probable Reason: This discrepancy arises due to the clipping in the DP-SGD algorithm, which affects the parameter updates during training.

Observation: For ERM SGD performs better than DP-SGD which is as expected. As the risk measure gets steeper the Differential **Probable Reason:**

7.2 Suboptimality and Worst Group Error for Amazon Dataset

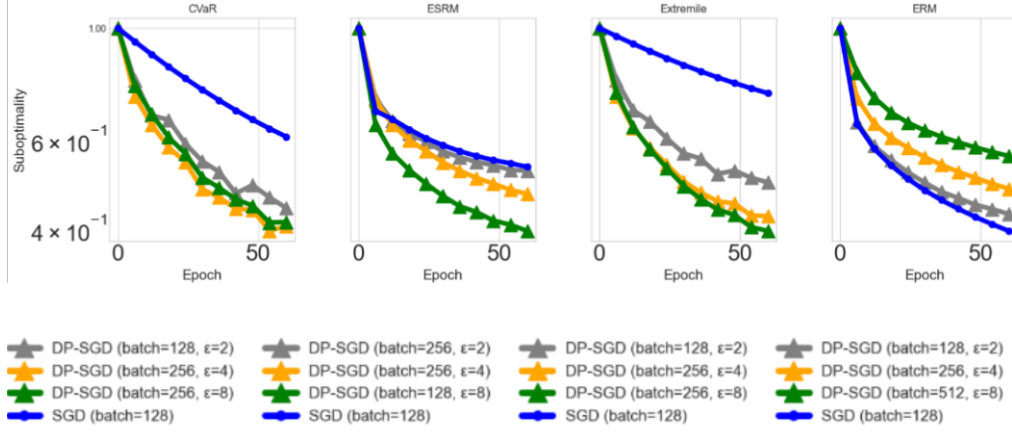


Figure 6: Suboptimality for Amazon Dataset

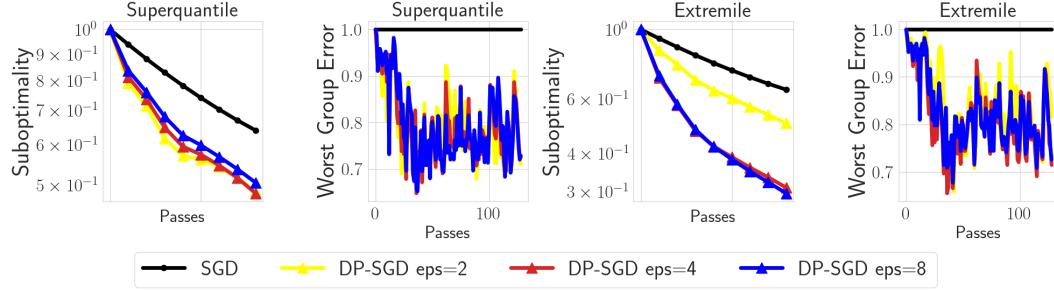


Figure 7: Suboptimality and Worst Group Error for Amazon Dataset

Observation: For ERM SGD performs better than DP-SGD which is as expected. As the risk measures get steeper the Differentially Private models perform better which is counter intuitive. The worst group error of DP-SGD models is better than SGD.

Probable Reason: The better optimal value for steeper spectral risks for DP-SGD models could be due to the regularization effect of adding noise to the gradients for differential privacy which might be aiding the optimization process. The clip norm for the gradients for all models is set to 1. However the dimension of the parameters is much higher in amazon dataset compared to acsincome. It could be possible that the clip norm required might be higher or different from the current value used. This discrepancy may leading to inconsistent results.

8 Code Reference

Here is the link to the Github repository for the experiments and implementation: [GitHub Repository](#)

References

- John M. Danskin. The theory of max-min, with applications. *SIAM Journal on Applied Mathematics*, 14(4):641–664, 1966. URL <http://www.jstor.org/stable/2946123>. Accessed: 27 Oct. 2024.
- H. Brendan McMahan. A survey of algorithms and analysis for adaptive online learning. *Google, Inc.*, 2013.

Ronak Mehta, Vincent Roulet, Krishna Pillutla, Lang Liu, and Zaid Harchaoui. Stochastic Optimization for Spectral Risk Measures. *Journal of Applied Mathematics and Optimization*, 2022.

Ronak Mehta, Vincent Roulet, Krishna Pillutla, and Zaid Harchaoui. Distributionally Robust Optimization with Bias and Variance Reduction. In *International Conference on Learning Representations*, 2024.

Jorge Nocedal and Stephen J. Wright. *Numerical Optimization*. Springer Series in Operations Research and Financial Engineering. Springer, 1999.

Shiori Sagawa, Pang Wei Koh, Tatsunori B. Hashimoto, and Percy Liang. Distributionally Robust Neural Networks For Group Shifts: On The Importance Of Regularization For Worst-Case Generalization. In *Advances in Neural Information Processing Systems*, 2020.

9 Appendix

9.1 Derivation of Gradient of Objective Function

Let us consider for a vector $u \in \mathbf{R}^n$,

$$g(u) = \max_{p \in \mathcal{P}} \left\{ \sum_i p_i u_i - \nu h(p) \right\}$$

Using **Danskin's Theorem** Danskin [1966]:

$$\begin{aligned} \nabla_u g_D(u) &= p_D^* = \arg \max_{p \in \mathcal{P}} \left(\sum_i p_i u_i - h(p) \right) \\ \Rightarrow \nabla_\theta g(l(\theta)) &= \nabla_\theta l(\theta) \cdot \nabla_u g(u) \Big|_{u=l(\theta)} \quad (\text{chain rule}) \\ &= \frac{\partial l(\theta)}{\partial \theta} \cdot \nabla_u g(u) \Big|_{u=l(\theta)} = \sum_{i=1}^n p_{D^i}^* \nabla_\theta l_i(\theta) \\ \Rightarrow \nabla R_D(\theta) &= \nabla_\theta \phi(\theta, p_D^*(\theta)) = p_D^{*T} \nabla_\theta l_D(\theta) \end{aligned}$$

9.2 Bounding the Difference Between p_D^* and $p_{D'}^*$

Let us consider:

$$\phi_1(p) = \nu h(p) - p^T l_D(\theta) \quad (\text{strongly convex})$$

$$\phi_2(p) = \nu h(p) - p^T l_{D'}(\theta) = \nu h(p) - p^T l_D(\theta) - p^T l_{diff}(\theta) = \phi_1(p) + (-p^T l_{diff}(\theta))$$

where we can take $\lambda = \frac{\nu}{p_{max}}$ where $p_{max} \geq p_i \forall i$. Hence, $\phi_2(p) = \phi_1(p) + \psi(p)$, where $\psi(p) = -p^T l_{diff}(\theta)$ (strongly convex). We can thus apply Lemma 7 from McMahan [2013] and conclude:

$$\|p_1^* - p_2^*\|_1 \leq \frac{\|\nabla \psi(p)\|_\infty}{\lambda} \leq \frac{B}{\lambda}$$

where $B = \max(\|l_{diff}(\theta)\|_2)$. Obtaining λ from: $\phi_2(p)$ is λ -strongly convex. If the entropy regularization term is defined as:

$$h(p) = \sum_{i=1}^n p_i \log p_i$$

then

$$\nabla^2 h(p) = \text{diag} \left(\frac{1}{p_1}, \frac{1}{p_2}, \dots, \frac{1}{p_n} \right)$$

and

$$\nabla^2 \phi_2(p) = \nu \text{diag} \left(\frac{1}{p_1}, \frac{1}{p_2}, \dots, \frac{1}{p_n} \right) \succeq \lambda I$$

where we can take $\lambda = \frac{\nu}{p_{max}}$ where $p_{max} \geq p_i \forall i$.

$$\|p_D^* - p_{D'}^*\|_1 \leq \frac{B}{\nu}$$